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SUPERSEDING
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MILITARY SPECIFICATION

MACHINE GUN, CALIBER .50: FIXED, M85

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of .50 caliber recoil-operated, air-cooled, link-belt fed, open-bolt firing, combat vehicle mounted machine gun with a quick-change barrel and an integral manual charger. The machine gun has two rates of fire, is fed from either the left hand or right hand side, and is fired by either a manual trigger or an electric solenoid.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

P-D-680	-	Dry Cleaning Solvent.
VV-L-800	-	Lubricating Oil, General Purposes, Preservative, (Water Displacing, Low Temperature).

Military

MIL-C-372	-	Cleaning Compound, Solvent (for Bore of Small Arms and Automatic Aircraft Weapons).
MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for.
MIL-P-14232	-	Parts, Equipment and Tools for Army Material, Packaging and Packing of.

FSC 1005

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MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of.
MIL-L-46000 - Lubricant, Semi-fluid, (Automatic Weapons).

STANDARDS

Federal
Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment.

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.

DRAWINGS

Rock Island Arsenal
F7793061 - Machine Gun, Caliber .50: Fixed, M85.
QE-G-DL7793061 - Quality Engineering Data List.

(Partdrawings referenced in this specification form a part of Drawing F7793061; inspection equipment drawings form a part of Data List DLQG 7793061.)

PUBLICATIONS

Rock Island Arsenal
P12002938 - Packaging Data Sheet for Machine Gun, Caliber .50: Fixed, M85.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1).

* 3.2 Materials, construction and design. The machine gun shall conform to the material, construction and design requirements specified herein, on Drawing F7793061, and in MIL-W-13855.

3.2.1 Accelerator assembly. The accelerator assembly shall readily assemble to the receiver, shall be held in position by the quick release pin, and shall readily disassemble from the receiver after removing the quick release pin. When the accelerator assembly is assembled to the receiver and the barrel is in the assembled position, the barrel latch block shall lock the barrel in the assembled position.

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3.2.1.1 Accelerator. The accelerator shall rotate about the pin through its full range of travel without binding and when released shall return to its original position by spring action.

3.2.1.2 Accelerator plunger. The accelerator plunger shall be capable of being depressed to allow the accelerator to rotate until contact is made with the legs of the accelerator housing, and when released the accelerator plunger shall return to its original position under spring action.

3.2.1.3 Cover detent. The cover detent shall be retained on the accelerator housing by the pin. The cover detent shall be capable of being depressed throughout the full range of travel without binding and when released shall return to its original position under spring action.

3.2.2 Barrel with flash suppressor. The barrel with flash suppressor shall slide without binding on the bearing surfaces of the barrel support, shall assemble to the barrel extension, shall rotate in the barrel extension to the locked position, and shall be retained in the locked position by the barrel latch block.

3.2.2.1 Flash suppressor. The flash suppressor shall be retained on the barrel by the pin as specified on the applicable drawing so that there shall be no relative movement.

3.2.2.2 Barrel. The barrel shall be free of cracks, seams, and mutilations. The bore and chamber shall be free of foreign matter, corrosion, pits, burs, sharp edges, bulges, and deformation. The chromium plating shall be free of nodules, anode burns, flaking and stripping. Scratches or marks, occurring in a chamber which otherwise meets the surface roughness requirements, shall be permitted provided they do not cause marks on the case of a high-pressure test cartridge fired in the chamber.

3.2.2.3 Barrel latch lock assembly. The barrel latch lock assembly shall be capable of being depressed throughout the full range of travel without binding and when released shall return to its original position under spring action. When the barrel latch lock is in the assembled position it shall prevent the barrel latch assembly from disengaging the barrel.

3.2.2.4 Barrel latch assembly. When the barrel latch lock assembly is depressed the barrel latch shall rotate to raise the barrel latch lock flush with or above the applicable surface of the accelerator housing, and when the barrel latch is released the barrel latch and barrel latch block shall return to its original positions under spring action.

3.2.3 Barrel extension assembly. The barrel extension assembly shall be capable of sliding without binding throughout the full range of travel in the slideways of the receiver.

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3.2.3.1 Barrel extension interlock. The barrel extension interlock shall be retained on the barrel extension assembly by the pin. The interlock shall be capable of being rotated throughout the full range of travel without binding, and when released shall return to its initial position under spring action.

3.2.3.2 Barrel return rod assemblies. The barrel return rod assemblies shall be retained in the barrel extension assembly by the straight pins. The barrel return rod shall be capable of being depressed flush with or below the applicable surface of the barrel extension assembly, and when released shall return to its original position under spring action.

3.2.4 Breech bolt. When the drive spring, feed and ejector assembly and sear assembly are removed from the machinegun, the breech bolt shall slide without binding in the slideways of the barrel extension except when the extractor engages the camming surface of the barrel extension.

3.2.4.1 Breechblock. When the breech bolt is removed from the machinegun, the breechblock shall move of its own weight in either direction through its full range of travel on the breechblock slide when the axis of the breech bolt is manually changed from the horizontal to the vertical position.

3.2.4.2 Breechblock slide. When the barrel extension assembly is removed from the machinegun and the breechblock is disassembled from the breechblock slide, the breechblock slide shall move in the slideways of the barrel extension assembly without binding throughout the full range of travel.

3.2.4.3 Right and left hand breechblock locks. The right and the left hand breechblock locks shall be cammed outward when the breechblock stops on the applicable surface of the barrel extension and the breechblock slide continues to move forward. The breechblock locks shall be retained within the breechbolt locks throughout the full height of the breechbolt locks.

3.2.4.4 Cartridge rammer. The cartridge rammer shall pivot about the pin without binding throughout the full range of travel and when released shall return to its original position by spring action.

3.2.4.5 Extractor. The extractor shall pivot without binding throughout its full range of travel and when released shall return to its original position under spring action. The extractor shall extract cartridge cases from the barrel chamber when the machinegun is fired or manually operated.

3.2.4.6 Cartridge case retainer. The cartridge case retainer shall pivot about its respective pin throughout the full range of travel without binding and when released shall return to its original position under spring action.

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3.2.4.7 Firing pin. The firing pin shall be retained in position on the breechblock slide and shall move in the breechblock without binding throughout the full range of travel.

3.2.5 Buffer sleeve. The buffer sleeve shall readily assemble to the receiver by sliding without binding in the slideways of the receiver.

3.2.6 Hand charger assembly. The hand charger assembly shall be capable of being assembled to either side of the receiver and when assembled to either side, shall be held in position by the slots of the receiver and charger detent and shall perform its function to engage and pull the breech bolt rearward for engagement with the bolt sear. After the breech bolt engages the bolt sear and the handle is released, the pin and pin retainer shall return to the forward position under spring action.

3.2.6.1 Charger detent assembly. The charger detent assembly shall be retained in position by the detent knob. The charger detent assembly shall be capable of being depressed flush with or below the applicable surface of the front stud and when released, shall return to its initial position under spring action.

3.2.7 Cover assembly. The cover assembly shall readily assemble to and disassemble from the receiver and when assembled, shall be retained in position by the quick release pin, shall rotate about the quick release pin, and when set in any one of the three indexed positions (approximately 30, 60, and 90 degrees with the axis of the bore), shall remain in the position set by action of the cover detent until manually reset. When the cover assembly is closed, the cover assembly shall be retained in this position by action of the latch.

3.2.7.1 Cover latch and latch lever assembly. The lever latch assembly shall be retained on the cover by the lock pin. The cover latch shall be capable of moving to a position flush with or below the applicable surface of the cover by rotating the latch lever assembly, and when the latch lever is released, the cover latch and latch lever assembly shall return to their original positions under spring action.

3.2.7.2 Cartridge guide pawl. The cartridge guide pawl shall be retained on the cover by the straight headed pin. The cartridge guide pawl shall rotate throughout its full range of travel without binding, and when released shall return to its original position under spring action.

3.2.7.3 Cartridge guide ramp. The cartridge guide ramp shall be retained on the cover by the quick release pin and the slot in the cover.

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3.2.7.4 Link stripper. The cartridge link stripper shall be retained on the cover by the two stripper retaining pins. The cartridge link stripper shall be capable of being depressed without binding throughout the full range of travel in the feed slide guide and when released shall return to its original position under spring action.

3.2.7.5 Feed slide assembly. The feed slide assembly shall be capable of being assembled for either right or left hand feed, when assembled for either feed position, the feed slide assembly shall slide in the slideways of the feed slide guide without binding throughout the full range of travel by rotating the feed lever. When the feed lever is released, the feed slide assembly shall return to its original position under spring action.

3.2.7.5.1 Feed slide guide. The feed slide guide shall be retained on the cover by the quick release pin and the slot in the cover.

3.2.7.5.2 Belt feed pawl. The belt feed pawl shall be retained on the feed slide by the straight headless pin. The belt feed pawl shall pivot throughout the full range of travel without binding and when released shall return to its original position under spring action.

3.2.7.5.3 Feed return housing assembly. The feed return housing assembly shall be capable of being assembled to the feed slide guide for either left or right hand feed and shall be retained in either position by the slot in the cover.

3.2.7.5.4 Cartridge stop pawl. The cartridge stop pawl shall be retained on the feed return housing by the pin. The cartridge stop pawl shall pivot about the pin throughout the full range of travel without binding and when released shall return to its original position under spring action.

3.2.7.6 Belt feed lever. The belt feed lever shall rotate about the pin and be retained on the pin by the feed slide guide. The belt feed lever shall mate with the feed actuator lever when the machinegun is assembled for either left or right hand feed and the breech bolt is in the forward position.

3.2.7.7 Spring guide pins. The spring guide pins shall mate with the holes in the feed slide assembly.

3.2.7.8 Lever pivot pin and tray pins. The lever pivot pin (7793152) and the tray pins (MS16555-40) shall be securely retained on the cover as specified on the applicable drawing so there shall be no relative movement.

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3.2.8 Back plate assembly. The back plate assembly shall readily assemble to the receiver and shall be retained on position by the back plate latch. When the latch lock and the back plate latch are depressed, the back plate assembly shall readily disassemble from the receiver. When the back plate assembly is in the assembled position in the receiver, the solenoid gap adjustment shall be in accordance with the applicable drawing.

3.2.8.1 Back plate latch and latch lock. The latch lock shall be capable of being depressed throughout its full range of travel without binding, and when released shall return to its original position under spring action. When the latch lock is depressed, the back plate latch shall rotate about the pin throughout its full range of travel without binding, and when released shall return to its original position under spring action. When the latch lock is in the assembled position, it shall prevent the back plate from being rotated sufficiently to disengage from the receiver.

3.2.8.2 Disconnector assembly. The disconnector assembly shall be retained on the back plate by the quick release pin and shall rotate about the quick release pin without binding throughout its full range of travel. The disconnector rollers shall be capable of rotating without binding. When the disconnector rollers are cammed to a rearward position by rotation of the cylinder, the disconnector shall prevent the armature from being depressed.

3.2.8.3 Rate selector. The rate selector shall be retained to the striker stud by the retainer. The rate selector shall be capable of being moved manually throughout its full range of travel, and when released shall return to its original position under spring action. The rate selector shall move to disengage from the slot of the back plate, rotate between the high and low rate position, and when set in either the high or low rate position, shall be retained in either the high or low rate position by engagement of the rate selector in the applicable slot of the back plate under spring action.

3.2.8.4 Striker assembly. When the rate selector is positioned at the letter "L", the striker shall be capable of being depressed throughout its full range of travel to rotate the cylinder, and when the striker is released it shall return to its original position under spring action. When the rate selector is positioned at the letter "H" and the breech bolt is pulled to the rearward position, the breech bolt shall not come in contact with the striker.

3.2.8.5 Time delay cylinder. The time delay cylinder shall be preloaded in accordance with the applicable drawing. When the rate selector is positioned at the letter "L", the cylinder shall be capable of being rotated throughout the full range of travel, and when released shall return, without binding, to its original position under spring action.

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3.2.9 Electrical. Unless otherwise specified, machineguns shall be capable of meeting the requirements specified herein using an input voltage of 24.0 ± 6.0 volts direct current (vdc).

3.2.9.1 Solenoid. The solenoid shall be securely retained on the back plate by the four socket head screws so that there shall be no relative movement. The socket head screws shall be tightened to the torque specified on the applicable drawing. The plunger assembly and armature shall move throughout the full range of travel without binding and shall return to their original positions under spring action. The dowel pins and the cable connector shall be securely retained in the solenoid so that there shall be no relative movement.

3.2.10 Trigger assembly. The trigger assembly shall be retained on the back plate by the quick release pin. The trigger assembly shall rotate about the quick release pin without binding throughout the full range of travel, causing the bolt sear to release the breech bolt (when seared), and when the trigger assembly is released it shall return to its original position under spring action.

3.2.10.1 Trigger extension assembly. The trigger extension assembly shall be retained on the back plate by the quick release pin. When the trigger extension assembly is pulled downward by the handle, the trigger extension shall rotate and shall cause the trigger assembly to rotate through the full range of travel.

3.2.11 Feed and ejector assembly. The feed and ejector assembly shall be capable of being assembled to either side of the receiver and shall be capable of being secured on either side of the receiver by the quick release pins. When the machinegun is fired or manually operated, the feed and ejector assembly shall cause the rounds to be properly positioned ahead of the belt retaining pawls of the feed and ejector assembly.

3.2.11.1 Feed actuator lever assembly. The feed actuator lever assembly shall be retained on the feed and ejector support by the pin. The feed actuator lever assembly shall rotate about the pin to compress the spring through the full range of travel without coming in contact with the ejector, and when released shall return to its original position under spring action.

3.2.11.2 Ejector. The ejector shall rotate about the pin throughout the full range of travel without binding, and when released shall return to its initial position under spring action. The ejector shall eject cartridge cases completely out of the machinegun when the machinegun is fired or manually operated.

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3.2.11.3 Retainer plunger. The retainer plunger shall be retained in position under spring action by the pin. The retainer plunger shall be capable of being depressed flush with or below the applicable surface of the feed and ejector support, and when released shall return to its original position under spring action.

3.2.11.4 Feed assembly retainer. The feed assembly retainer shall slide throughout the full range of travel in the applicable slot on either side of the feed and ejector support to depress the retainer plunger and spring, and when released shall return to its original position under spring action.

3.2.11.5 Belt retaining pawls. The belt retaining pawls shall be capable of being assembled on either side of the feed and ejector support and shall be retained by the pawl retaining pin. When assembled on either side of the feed and ejector support, the belt retaining pawls shall rotate about the pawl retaining pin throughout the full range of travel, and when released shall return to their original position under spring action.

3.2.11.6 Feed assembly actuator. When the feed assembly retainer and the feed actuator lever are depressed through their full range of travel, the feed assembly actuator shall slide without binding throughout the full range of travel in the slideways of the feed and ejector support.

3.2.11.6.1 Feed plunger assembly. The feed plunger assembly shall be capable of being depressed throughout the full range of travel without binding, and when released shall return to its original position under spring action.

3.2.12 Guide rod. The guide rod shall readily assemble to the receiver by mating with the hole in the front of the receiver and by rotating the guide rod in the groove at the rear of the receiver until the tab mates with the locking slot of the receiver. When the back plate assembly is assembled to or disassembled from the receiver, the tab of the guide rod shall not disengage from the locking slot of the receiver.

3.2.13 Feed tray assembly. The feed tray assembly shall rotate about the quick release pin throughout the full range of travel without binding. When the cover assembly is in the open position, the feed tray assembly shall remain in the open position by binding with the ramp cartridge guide. When the feed tray assembly and the cover assembly are in the closed position, the positioning holes of the feed tray shall mate with the positioning pins of the cover assembly.

3.2.13.1 Link guide assembly. The link guide assembly shall be capable of being assembled on either side of the feed tray and lock in the proper position by action of the link guide plunger.

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3.2.13.1.1 Link guide plungers. The link guide plungers shall be capable of being depressed flush with or below the applicable surfaces of the link guide, and when released shall return to their initial positions under spring action.

3.2.14 Quick release pins. All quick release pins shall perform their retaining function and shall be secured in position by spring action of the balls.

3.2.15 Receiver assembly. The slideways of the receiver assembly shall be smooth and free of burs. All riveted parts shall be tight, undamaged, and without relative movement.

3.2.16 Safety assembly. The safety assembly shall be capable of being secured to either side of the receiver and when assembled to either side shall be securely retained in place by the screw so that there shall be no relative movement of the safety housing. The pin shall be securely retained in the safety plunger so that there shall be no relative movement of the pin in the plunger. When the safety assembly is assembled to either side of the receiver, the pin shall move manually between the safe and fire positions and shall remain in the position set under spring action until reset manually. When the safety assembly is assembled to either side of the receiver and the pin is positioned at the letter "F", the safety plunger shall allow the bolt sear, when rotated, to release from the breech bolt. When the safety assembly is assembled to either side of the receiver and the pin is positioned at the letter "S", the safety plunger shall prevent the bolt sear from releasing the breech bolt.

3.2.17 Sear assembly. The sear assembly shall readily assemble on the receiver and when assembled shall be retained on the proper position by the sear detent. The sear assembly shall readily disassemble from the receiver by depressing the sear detent and sliding the sear assembly rearward out of the receiver.

3.2.17.1 Bolt sear. The bolt sear shall be retained to the sear assembly by the pin. The bolt sear shall rotate about the pin throughout the full range of travel without binding when activated by the bolt sear lever and shall return to its initial position when the bolt sear lever is released. The bolt sear shall engage the bolt sear engagement surface of the breech bolt and shall hold the breech bolt in the seared position. When the bolt sear is rotated (by activating the bolt sear lever) the bolt sear shall disengage from the bolt sear engagement surface of the breech bolt.

3.2.17.2 Bolt sear lever. The bolt sear lever shall be retained on the sear assembly by the pin. The bolt sear lever shall rotate about the pin throughout the full range of travel without binding.

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3.2.17.3 Sear detent. The sear detent shall be capable of being depressed flush with or below the applicable surface of the sear block and when released, shall return to its initial position under spring action.

3.3 Performance characteristics.

3.3.1 Firing pin protrusion. With the breech bolt in the fully locked position, the firing pin protrusion in the assembled machinegun shall be not less than 0.021 inch nor more than 0.047 inch. Testing shall be as specified in 4.5.3.2.

3.3.2 Sear release. The force required to release the breech bolt from the bolt sear shall be not less than 30 pounds (lbs) nor more than 50 lbs applied to the trigger in a direction parallel to the axis of the machinegun. Testing shall be as specified in 4.5.3.2.

* 3.3.3 Breech bolt gap. Before acceptance firing, the breech bolt gap (gap between the barrel extension and the breech bolt locks) shall be not more than 0.018 inch. After acceptance firing the breech bolt gap shall be not more than 0.023 inch. Testing shall be as specified in 4.5.3.3.

* 3.3.4 Headspace. The headspace shall be not less than 3.118 inches nor more than 3.133 inches when measured to the 0.636 diameter datum on the first shoulder of the chamber. Testing shall be as specified in 4.5.3.4.

* 3.3.5 High-pressure resistance. The machinegun shall be capable of withstanding the firing of one round of Government standard .50 caliber, M1 high pressure test cartridge. Parts shall be free of cracks, seams, and other injurous defects after proof firing. Testing shall be as specified in 4.5.3.5.

3.3.6 Accuracy. Twenty rounds of a 20 round burst, fired at low rate from the machinegun at a range of 1,000 inches shall be not more than 4.0 inches figure of merit. Testing shall be as specified in 4.5.3.6.

3.3.7 Functioning firing. The machinegun shall operate without malfunctions or unserviceable parts, shall have a high and a low cyclic rate of fire of 800 to 950 and 400 to 500 rounds per minute respectively, and shall withstand a belt pull of 20 lb. Testing shall be as specified in 4.5.3.7.

3.3.8 Endurance. Machineguns shall be capable of firing 10,000 rounds of M33 ball cartridges, using M15A2 links, with not more than the number of malfunctions and unserviceable parts allowed in Table I. The average high and average low cyclic rate of fire for the entire test shall be within the limits specified in 3.3.7. Testing shall be as specified in 4.5.3.8.

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* 3.4 Interchangeability. Unless otherwise specified on the drawings all parts shall be interchangeable. Testing shall be as specified in 4.5.3.9.

Table I. Malfunctions and Unserviceable Parts

Code	Malfunctions[1]	Malfunctions permitted in 10,000 rounds
FX	Failure to extract	2
FJ	Failure to eject	1
LB	Light blow	1
FFCR	Failure to feed or strip (correctable by recharging)	5
FFNC	Failure to feed or strip (not correctable by recharging)	2
	Total malfunctions - above malfunctions combined	8

Code	Unserviceable Parts[1]	Unserviceable parts permitted in 10,000 rounds	
		First 5,000 rounds	Second 5,000 rounds
EX	Extractor	0	1
GR	Guide rod	0	1
EXS	Extractor spring	0	1
RAS	Rammer spring	0	1
CRS	Cartridges retainer spring	0	1
OP	Other parts[2]	0	1

[1]When malfunctions are traceable to particular parts, it is permissible to replace such parts and record them as unserviceable, subject to limitations of Table I. When it is definitely established by the Government representative that previously recorded malfunctions are attributable to an unserviceable part, such malfunctions shall not be counted against the machine gun being tested, provided that they occurred not more than 200 rounds prior to replacement of the unserviceable part. These 200 rounds shall have been fired with the unserviceable part. However, such malfunctions shall remain recorded and properly identified. An unserviceable part isone that causes malfunctions or impairs the safety of the weapon. Malfunctions attributable to links and ammunition shall not be counted against the machine gun, however, such malfunctions shall be recorded. (All malfunctions which are not counted against the machine gun,

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(i.e., attributed to an unserviceable part, Government furnished equipment, etc.) shall be supported by documented objective evidence that the malfunction was caused by a defective condition of an unserviceable part, Government furnished equipment, ammunition, etc.).

[2]One unserviceable part other than those specified shall be allowed if in the judgment of the Government representative the failure does not represent an unsafe or represent an unsafe or defective condition which is prevalent throughout the lot of items involved.

3.5 Marking. The machine gun shall be marked in accordance with MIL-W-13855. Each machinegun shall be identified by a serial number assigned by the procuring activity (see 6.1).

3.6 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are as follows:

1. First Article inspection 4.4.
2. Quality conformance inspection 4.5

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.1). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein and such other inspection as necessary to determine that all requirements of the contract have been met.

4.5 Quality conformance inspection.

4.5.1 Inspection lot.

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4.5.1.1 Machineguns. The number of machineguns in an inspection lot shall be either 200 or one month's production, whichever is smaller. Machineguns shall be assembled from lots of component parts that have met all inspection requirements specified herein. Endurance test lot size shall be as specified in 4.5.3.8.1.

4.5.2 Examination.

4.5.2.1 Component parts and concurrent repair parts. Examination of component parts and concurrent repair parts shall be performed in accordance with the criteria specified in the contract (see 6.1). The contractor's examination of these parts shall be accomplished prior to their assembly into the end item or submission for acceptance as repair parts.

* 4.5.2.2 Machineguns. Visually and manually examine each machinegun in accordance with the following. Each step in the examination shall include a visual examination for proper cleaning and presence of the specified protective coating and to determine the general quality, completeness of manufacture, assembly, and workmanship. The examination provisions shall be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Reinspections of these characteristics on the completed product is not required provided assurance exists that the characteristic has not been changed, degraded or damaged by subsequent manufacturing, assembly or handling and that adequate inspection records are maintained. Machineguns failing to meet the requirements shall be rejected.

4.5.2.2.1 Safety assembly. Visually and manually examine the safety assembly to determine compliance with 3.2.16.

4.5.2.2.2 Hand charger assembly.

a. Manually examine the hand charger assembly to determine compliance with 3.2.6 and 3.2.6.1.
b. Visually and manually examine to assure that the handle is retained in position on the chain by the two expanded links (see Drawing B11010488).
c. Visually examine the hand charger assembly to assure freedom from burs, cracks, sharp edges, and deformations.

4.5.2.2.3 Cover assembly.

a. Visually and manually examine the cover assembly to determine compliance with 3.2.7 through 3.2.7.8.
b. Visually examine components of the cover assembly to assure freedom from burs and deformations.
c. Visually and manually examine the quick release pin to assure the two detent balls are retained in the pin.

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4.5.2.2.4 Feed tray assembly.

a. Visually and manually examine the feed tray assembly to determine compliance with 3.2.13 through 3.2.13.1.1.
b. Visually examine the feed tray assembly to assure freedom from burs, cracks, and deformations.

4.5.2.2.5 Trigger extension assembly.

a. Visually and manually examine the trigger extension assembly to determine compliance with 3.2.10.1.
b. Visually and manually examine to assure that the handle is retained in position on the chain by the two expanded links (see Drawing C11010486.)
c. Visually and manually examine chain slot on trigger extension to assure proper width (see Drawing C11686373) and to assure secure retention of chain to the trigger extension.

4.5.2.2.6 Back plate assembly.

a. Manually examine back plate assembly for ability to be readily disassembled (see 3.2.8).
b. Manually examine the back plate latch and latch lock to determine compliance with 3.2.8.1.
c. Manually examine the rate selector to determine compliance with 3.2.8.3.
d. Visually examine the low and high rate markings on the back plate to assure legibility.
e. Visually and manually examine the back plate subassembly to assure secure welding of plate (for alternate design only, see Drawing C7791385).
f. Manually examine the time delay cylinder to determine compliance with 3.2.8.5.
g. Manually examine the trigger assembly to determine compliance 3.2.10.
h. Disassemble the trigger assembly and manually examine the spring pin retaining the trigger cap to the trigger for proper assembly and secure retention (see Drawing C11010179).
i. Disassemble the solenoid and disconnector assembly and manually examine the plunger assembly and armature for freedom of movement (see 3.2.9.1).
j. Manually examine the disconnector rollers for freedom of rotation (see 3.2.8.2).
k. Visually and manually examine the pins retaining the rollers to assure secure staking (see Drawing C7793088).
l. Visually examine the striker assembly to assure secure flaring of the striker retaining pin and to assure that the pin is flush with or below the striker surface (see Drawing B7791361).
m. Manually examine the solenoid to assure dowel pins and the cable connector are secure.
n. Visually examine all parts of the back plate assembly to assure freedom from burs, cracks and deformations.

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o. Manually examine the quick release pin to assure the two detent balls are retained in the pin.
p. Reassemble the disconnector assembly and manually examine to assure freedom of rotation on the quick release pin (see 3.2.8.2).
q. Reassemble the trigger assembly and the trigger extension assembly and manually examine for proper function (see 3.2.10 and 3.2.10.1).
r. Manually examine the striker assembly with the rate selector positioned at the letter "L" to determine compliance with 3.2.8.4.
s. Reassemble solenoid and manually examine for secure retention (see 3.2.9.1).
t. Manually rotate the cylinder so that the disconnector rollers are cammed to the rearward position and examine to assure that the disconnector prevents the armature from being depressed (see 3.2.8,2).

4.5.2.2.7 Guide rod.

a. Visually and manually examine the guide rod to determine compliance with 3.2.12.
b. Disassemble springs, guide rod, and buffer sleeve.
c. Visually and manually examine the ends of spring (7791254) for secure brazing and to assure freedom from loose end strands and deformations.
d. Visually examine spring (7790739) to assure freedom from broken ends, cracks, burs, and deformations.

4.5.2.2.8 Buffer sleeve. Visually and manually examine the buffer sleeve to determine compliance with 3.2.5.

4.5.2.2.9 Barrel with flash suppressor.

a. Visually and manually examine the barrel with flash suppressor to determine compliance with 3.2.2, 3.2.2.1 and 3.2.2.2.
b. Visually examine the extension lugs and barrel bearing surfaces to assure freedom from burs, cracks, and mutilations.

4.5.2.2.10 Feed and ejector assembly.

a. Visually and manually examine the feed and ejector assembly to determine compliance with 3.2.11 through 3.2.11.2 and 3.2.11.6.
b. Manually examine the belt retaining pawls to determine compliance with 3.2.11.5.
c. Disassemble the feed actuator assembly and visually and manually examine the retainer plunger and feed assembly retainer to determine compliance with 3.2.11.3 and 3.2.11.4.
d. Manually examine the feed plunger assembly to determine compliance with 3.2.11.6.1.

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e. Visually examine the quick release pins to assure the two detent balls are retained in the pins.
f. Visually examine the legs of the ejector to assure proper alinement and to assure freedom from deformation.
g. Visually examine all components to assure freedom from burs, cracks, and deformations.
h. Assemble all components of the feed and ejector assembly examining for proper function.

4.5.2.2.11 Sear assembly.

a. Visually and manually examine the sear assembly to determine compliance with 3.2.17 through 3.2.17.3.
b. Visually examine the sear assembly to assure freedom from burs, cracks, and deformations.

4.5.2.2.12 Barrel extension assembly.

a. Visually and manually examine the barrel extension assembly to determine compliance with 3.2.3, 3.2.3.1 and 3.2.3.2.
b. Visually and manually examine the barrel stop pin to assure it is securely retained in the barrel extension.
c. Visually examine the barrel extension to assure freedom from burs, cracks, deformations, and defective chrome plating.
d. Visually examine the barrel extension assembly to assure presence of proof and magnetic particle inspection mark.

4.5.2.2.13 Breech bolt.

a. Visually and manually examine the breech bolt to determine compliance with 3.2.4 and 3.2.4.1.
b. Disassemble breechblock from breechblock slide and manually examine to determine compliance with 3.2.4.2.
c. Manually examine the right and left hand breechblock locks to determine compliance with 3.2.4.3.
d. Visually examine components to assure freedom from burs, cracks, deformations, and defective chrome plating.
e. Visually examine breech bolt to assure presence of proof and magnetic particle inspection marks.
f. Reassemble components and visually and manually examine to determine compliance with 3.2.4.3 through 3.2.4.7.

4.5.2.2.14 Accelerator assembly.

a. Visually and manually examine the accelerator assembly to determine compliance with 3.2.1 through 3.2.1.3 and 3.2.2.3, 3.2.2.4.
b. Visually examine barrel latch lock retaining pin for centrality (see Drawing C7790980).

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c. Visually examine the accelerator assembly to assure freedom from burs, cracks, and deformation.
d. Visually examine the quick release pins to assure the two detent balls are retained in the pins.

4.5.2.2.15 Receiver assembly. Visually and manually examine the receiver assembly to determine compliance with 3.2.15.

4.5.2.2.16 Machine gun assembly.

a. Reassemble all parts of the machine gun for left hand feed examining for ability of parts to be readily assembled (see 3.2.1, 3.2.2.2, 3.2.6 through 3.2.8.5 and 3.2.9.1, 3.2.10, 3.2.11, 3.2.12 and 3.2.17) and to assure the quick release pins perform their retaining function (see 3.2.14).
b. Hand function the machine gun using five dummy rounds to assure proper operation.
c. Manually examine function of safety (see 3.2.16).
d. Manually examine for solenoid gap adjustment (see Drawing F7793061.
e. Visually examine the safety assembly to assure it is assembled on the feed side of the receiver.

4.5.2.2.17 Rejected machine guns. Rejected machine guns shall be corrected and the corrective action taken shall be reexamined for all defects listed in the applicable groups in 4.5.2.2.

4.5.2.2.17.1 When any of the following parts are replaced to correct a nonconforming machine gun, the machine gun shall be resubmitted to both or either of the following tests as indicated by asterisks:

		Function Firing Test	Target & Accuracy Test
1.	Safety Assembly	*	
2.	Hand Charger Assembly	*	
3.	Cover Assembly	*	
4.	Feed Tray Assembly	*	
5.	Trigger Extension Assembly	*	
6.	Back Plate Assembly	*	
7.	Guide Rod	*	
8.	Barrel w/Flash Suppressor	*	*
9.	Feed and Ejector Assembly	*	
10.	Sear Assembly	*	
11.	Barrel Extension Assembly	*	
12.	Breech Bolt	*	
13.	Acceleration Assembly	*	
14.	Receiver Assembly	*	*

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4.5.3 Testing.

4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete machine gun. Failure of machine guns to meet the test requirements specified herein shall be cause for rejection of the machine gun or the inspection lot of machine guns. Acceptance of machine guns shall be deferred and the contractor shall accomplish as applicable the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable productions processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine machine guns, partially assembled machine guns and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned machine guns for retest.

4.5.3.2 Firing pin protrusion and sear release testing. A sample of 10 machine guns selected from each inspection lot, shall be tested for firing pin protrusion (see 3.3.1) and sear release (see 3.3.2) using the test methods specified in 4.6.1 and 4.6.2 respectively.

4.5.3.3 Breech bolt gap testing. Each machine gun shall be tested for breech bolt gap (see 3.3.3) using the test method specified in 4.6.3.

4.5.3.4 Headspace testing. Each machine gun shall be tested for the headspace requirement (see 3.3.4) using the test method specified in 4.6.4.

4.5.3.5 High-pressure resistance testing. Each machine gun shall be tested for high-pressure resistance (see 3.3.5) using the test method specified in 4.6.5.

4.5.3.6 Accuracy testing. Each machine gun shall be tested for the accuracy requirements (see 3.3.6) using the test method specified in 4.6.6 until 200 consecutive machine guns meet the requirements. When 200 consecutive machine guns have met the requirements, the contractor shall test a sample taken from each inspection lot in accordance with inspection level S4 of MIL-STD-105. Acceptance shall be based on an AQL 1.0 percent

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defective. If an inspection lot is rejected, 100 percent testing shall be reinstated until 200 consecutive machine guns meet the requirements, at which time the contractor shall return to the above sampling plan. Machine guns failing to meet the accuracy requirement during 100 percent testing shall be rejected.

4.5.3.7 Function firing testing. Each machine gun shall be tested for function firing (see 3.3.7) using the test methods specified in 4.6.7. After 100 machine guns are function fired as specified in Table III, and they meet the requirements, subsequent function testing shall be based on the reduced firing schedule. In the event of a failure to meet the requirements, the contractor shall resume firing at the greater number of rounds. Changes in the firing schedule shall be subject to Government approval based on test results and corrective actions taken

4.5.3.8 Endurance testing.

4.5.3.8.1 Lot size. The first five endurance test lots shall each consist of 200 machine guns or a months' production, whichever is smaller. When five successive lots meet the endurance requirements, the lot size shall be increased to 500 machine guns or a months' production whichever is smaller. When five successive lots of the increased size have met the endurance requirements, the lot size shall be further increased to 1,000 machine guns or a months' production, whichever is smaller. If rejection of a lot occurs at any time, the next smaller test lot size criteria shall be reinstated and the above procedure repeated to the larger lot size.

4.5.3.8.2 Procedure. One machine gun selected at random from each endurance lot shall be tested for endurance (see 3.3.8) using the test method specified in 4.6.8. The contractor shall provide replacement parts as required to complete the test. If the endurance requirements are not met, the represented lot shall be rejected.

4.5.3.9 Interchangeability testing.

4.5.3.9.1 In plant.

* 4.5.3.9.1.1 Machine guns. A sample of ten machine guns selected at random from each inspection lot, shall be tested for interchangeability (see 3.4) using the test method specified in 4.6.9.1.1. Machine guns taken for interchangeability testing shall have been found satisfactory in all other examinations and tests. Test frequency may be reduced, on approval of the contracting officer, to not less than one test of ten machine guns each month when a record of consistency has been established. The machine gun shall be inspected for and shall comply with the requirements for firing pin protrusion, sear release, breech bolt gap, headspace, accuracy and functioning firing requirements after interchange of parts using the inspection methods, specified in 4.6.1, 4.6.2, 4.6.3, 4.6.4, 4.6.6 and 4.6.7 respectively. At the discretion of the procuring activity, failure of the

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interchangeability test shall cause rejection or retest of the represented lot, the retest may be allowed without reconditioning the lot. Failure in the retest shall cause rejection of the represented lot subject to reconditioning and further test as a reconditioned lot. A sample of 20 machine guns from each retest or reconditioned lot shall be tested using the same procedure described above.

* 4.5.3.9.1.2 Concurrent repair parts. At least two parts from each inspection lot of concurrent repair parts shall be subjected to the interchangeability test specified in 4.6.9.1.2. Failure of any part to meet the requirements shall be cause for rejection of the represented lot of parts subject to reconditioning and further test as a reconditioned lot. A sample of double the number of parts used in the original test shall be tested from each reconditioned lot using the test method specified in 4.6.9.1.2.

4.5.3.9.2 Interplant. When machine guns are manufactured concurrently by more than one contractor, each contractor shall forward monthly six machine guns for the interplant interchangeability test specified in 4.6.9.2 to testing agency specified in the contract (see 6.1). They will be informed of the results of the test which indicates failure of the machine guns to meet prescribed requirements.

* 4.5.3.10 Certification. Unless otherwise specified, the contractor shall furnish the Government representative with certified statements that each inspection lot of machine guns conforms to the materials and processes specified on the applicable drawings and specifications.

4.5.3.11 Component parts and concurrent repair parts testing. Raw material testing, part testing, and certification shall be performed in accordance with the criteria specified in the contract (see 6.1). This will include chemical analysis and physical tests of materials, and tests of protective finish, heat treatment, bonding, and function of parts as applicable. These tests shall be accomplished prior to assembly of parts into the end item.

4.5.4 Packaging examination and testing. Unless otherwise specified (see 6.1), the packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.5 Inspection equipment.

4.5.5.1 Unless otherwise specified in the procurement documents (see 6.1), responsibilities for acquisition, maintenance, and disposition of measuring and testing equipment prescribed on lists contained on the Quality Engineering Data List QE-G-DL7793061, and for all other inspection equipment required to perform inspection required by applicable specifications, shall be in accordance with MIL-I-45607.

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4.5.5.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.5.5.3 Power supply. A power supply capable of supplying 15 amperes over a range of 18.0 to 30.0 vdc shall be used to operate the solenoid.

4.6 Test methods.

4.6.1 Firing pin protrusion test. Assembled machine guns shall be tested for firing pin protrusion requirement (see 3.3.1) using inspection equipment in accordance with Drawing F11016381. The machine gun shall be placed in a horizontal position (approximately), and the cover assembly and the feed tray assembly opened. The breech bolt shall be retracted to the seared position. The spring loaded plug shall then be inserted in the barrel chamber and the breech bolt released from the bolt sear and manually returned to the fully locked position. The indicator portion of the gage shall be set to zero using the setting check as specified on the applicable drawing and then inserted in the barrel from the muzzle end until it contacts the spring loaded plug. The indicator shall then be read to determine if the assembled machine gun firing pin protrusion is within the specified requirement.

4.6.2 Sear release test. Machine guns shall be tested for sear release requirement (see 3.3.2), using the inspection equipment in accordance with Drawing F7797499. The machine gun shall be placed in a horizontal position (approximately) and the breech bolt retracted to the seared position. The force indicator portion of the gage shall be in the retracted position and the telltale hand set to zero. The gage shall be placed on the cover and retained in position by releasing the side knobs. The handle of the gage shall be manually moved slowly and steadily toward the muzzle end of the machine gun until the breech bolt is released from the bolt sear. The handle shall then be manually returned to its original position. The telltale hand of the indicator shall be checked to determine if the sear release requirement has been met.

4.6.3 Breech bolt gap test. Machine guns shall be tested for the breech bolt gap requirement (see 3.3.3) using the inspection equipment in accordance with Drawing C11015617. The machine gun shall be placed in a horizontal position (approximately) with the cover assembly and the feed tray assembly opened and the breechlock, breechblock slide, and breechblock locks (left and right) fully forward. The resultant gaps between the breechblock locks (left and right) and the barrel extension shall be checked with the gage to determine if the breech bolt gap is within the specified requirements prior to and after acceptance firing.

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4.6.4 Headspace test. Machine guns shall be tested for headspace requirement (see 3.3.4) using inspection equipment in accordance with Drawing F7797438. The machine gun shall be placed in a horizontal position (approximately), and the cover assembly and the feed tray assembly opened. The breech bolt shall be retracted to the seared position. The spring loaded plug shall then be inserted in the barrel chamber and the breech bolt released from the bolt sear and manually returned to the fully locked position. The indicator portion of the gage shall be set to zero using the setting as specified on the applicable drawing and then inserted in the barrel from the muzzle end until it contacts the spring loaded plug. The indicator shall then be read to determine if the headspace is within the specified requirement.

4.6.5 High-pressure resistance test. Machine guns shall be tested for high-pressure resistance requirement (see 3.3.5) by firing one high-pressure test cartridge in each machine gun. Machine guns shall be tested using the firing fixture conforming to Drawing F7273976 and using a safety shield conforming to Drawing F7273975. Breech bolts, barrel extension subassemblies, and barrels to be assembled into production machine guns shall be proof fired only as part of the completed machine guns magnetic particle inspected, and marked in accordance with the applicable drawing. Breech bolts, barrel extension subassemblies, and barrels to be used as repair parts shall be proof fired independently of the production machine guns, magnetic particle inspected, and marked in accordance with the applicable drawing. After proof firing, machine guns shall be examined for cracks, deformations, and other evidence of damage, and cartridge cases shall be visually examined for bulges, splits, rings, and other defects caused by defective barrels. Proof marks shall be applied as indicated on the applicable drawings on machine guns that have passed this test.

4.6.6 Accuracy test.

4.6.6.1 The machine guns shall be tested for accuracy requirement (see 3.3.6) using the firing fixture conforming to Drawing F7273891. Machine guns may be fired at a range greater than 1,000 inches with suitable correlation in the accuracy requirement as indicated in Table II and as approved by the procuring activity.

4.6.6.2 At the beginning of each day's firing, the first machine gun shall be boresighted to bring the point of aim to the center of the target. The boresight shall be in accordance with Drawing F7797501. The firing of a 20-round burst for warmup is allowable before the machine gun is fired for accuracy. A 20-round burst shall be fired at low rate and the target checked for the accuracy requirement. The dispersion shall be computed by "figure of merit". The figure of merit value shall be recorded. The ball ammunition and links used in this test shall be in accordance with 3.3.8.

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4.6.6.2.1 Rejected machine guns. Machine guns rejected for failure to meet the accuracy requirements shall be corrected, and the corrective action shall be recorded.

4.6.6.2.1.1 Corrected machine guns shall be retested by firing 20 round bursts, low rate, at each of two targets. The targets shall be checked to determine that machine guns meet the accuracy requirements.

Table II. Accuracy Requirement Correlation

Range	Figure of merit[1]
1,000 inches	4.0 inches
50 yards	7.2 inches
100 yards	14.4 inches

[1]Figure of merit equals one-half the sum of the extreme horizontal dispersion and extreme vertical dispersion. Measurements shall be made between centers or between corresponding edges of the applicable shot holes.

4.6.7 Functioning firing test.

4.6.7.1 Machine guns shall be tested for functioning firing requirement (see 3.3.7) in accordance with the schedule specified in Table III. The machine guns shall be held in a firing fixture conforming to Drawing F7273976.

4.6.7.2 Machine guns may be fired using 40 and 50 or 75 and 85 round belts of ball ammunition (as applicable). Dummy-inert cartridges shall be used to separate each series of firing. Caution should be employed in the use of the dummy-inert cartridges to assure that both the case and projectile of the dummy-inert cartridge are ejected from the weapon during the hand charging operation.

4.6.7.3 The voltage for operation of the solenoid shall be 24 ± 6 vdc.

4.6.7.4 Firing shall be intentionally stopped (by releasing the manual trigger or discontinuing the electric current to the solenoid, as applicable) at least two times during the interrupted burst firing. The machine gun shall be checked to assure that the firing stops immediately when the manual trigger is released or the electric current to the solenoid is discontinued.

4.6.7.5 The cyclic rate of fire shall be measured and recorded during one continuous burst at the high rate setting and during one continuous burst at the low rate setting using a timer gage conforming to Drawing D7273920 or equivalent contractor designed, Government approved.

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4.6.7.6 Ten live rounds with three dummy-inert rounds attached shall be fired in a continuous burst using the firing fixture with the flexible chuting removed and the wheel attached. The belt pull weight specified in 3.3.7 shall be attached to the dummy-inert rounds by means of the cable, or other contractor designed Government approved test fixture.

Table III. Functioning Firing Schedule (see 4.6.7.1)

Feed direction and method of firing	Rate of fire	Number of rounds (see 4.5.3.7) First 100 machine guns	Reduced firing schedule	Type of firing
Right hand feed using the solenoid (see 4.6.7.3)	High	25	15	Interrupted burst (see 4.6.7.4)
	High	50	25	Continuous burst (see 4.6.7.5)
	Low	25	15	Interrupted burst (see 4.6.7.4)
	Low	50	25	Continuous burst (see 4.6.7.5)
	Low	10	10	Belt pull (see 4.6.7.6)
Left hand feed using the manual trigger	High	25	15	Interrupted burst (see 4.6.7.4)
	High	50	25	Continuous burst (see 4.6.7.5)
	Low	25	15	Interrupted burst (see 4.6.7.4)
	Low	50	25	Continuous burst (see 4.6.7.5)
	Low	10	10	Belt pull (see 4.6.7.6)

4.6.8 Endurance test.

4.6.8.1 Testing of machine guns for endurance requirement (see 3.3.8) shall be accomplished with the machine guns held in a firing fixture conforming to Drawing F7273976.

* 4.6.8.2 The machine gun being tested for endurance shall be assembled for left-hand feed at the start of the test, and the direction of feed shall be changed after firing 3,240, 6,480 and 9,000 rounds. The first 990 rounds of the test shall be electrically fired, the second 990 rounds shall be fired using manual trigger, and remainder of the firing test shall be electrically fired. The first 7,470 rounds of the test shall be fired at a low rate and remainder of the test shall be fired at a high rate. Three barrel assemblies with flash suppressors shall be used during this test. The barrel assembly shall be replaced after each 3,510 rounds of firing.

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* 4.6.8.3 The firing schedule for the first 9,000 rounds of the endurance test shall be fired as follows: 90 rounds in an interrupted burst followed by 90 rounds in a continuous burst, using 90 round belts. During the firing of the first 90 round continuous burst of each 1,620 rounds, the cyclic rate of fire shall be measured and recorded using the timer gage conforming to 4.6.7.5. The last 1,000 rounds of the endurance test shall be in accordance with 4.6.8.4. The first 1,620 rounds of the test may be fired with lubricant conforming to either MIL-L-46000 or VV-L-800. The machine gun shall be lubricated with lubricant conforming to MIL-L-46000 and gymnasticated prior to the start of each days firing. After each 270 rounds fired, the machine gun shall be allowed to cool at least five (5) minutes by forced air. After 1,620, 4,860 and 7,830 rounds are fired, coat the friction bearing surfaces with lubricant conforming to MIL-L-46000 without disassembling the weapon. After 3,240, 6,480, and 9,000 rounds fired, and at the end of each day's firing, the machine gun shall be cleaned with dry cleaning solvent conforming to P-D-680, dried and lubricated with lubricant conforming to MIL-L-46000. Parts exposed to powder residue may be cleaned with rifle bore cleaner conforming to MIL-C-372 prior to cleaning, drying and lubricating as specified above. Weapon's solenoid shall not be cleaned with dry cleaning solvent, rifle bore cleaner or lubricated.

4.6.8.3.1 Firing shall be intentionally stopped (by releasing the manual trigger or discontinuing the electrical current to the solenoid, as applicable) at least five times during the interrupted burst firing. The machine gun shall be checked to assure that the firing stops immediately when the trigger is released or the current to the solenoid is discontinued.

4.6.8.4 The firing schedule for the last 1,000 rounds shall be at a rate of approximately 150 rounds per minute in 2-second bursts (approximately 30 rounds) with a 10-second delay between bursts. The machine gun shall be cooled for 10 minutes by forced air after each 500 rounds.

4.6.8.5 Ball ammunition and links used in this test shall be in accordance with 3.3.8.

4.6.8.6 The safety precautions listed below should be followed in handling a "hot gun" when a stoppage occurs. (A "hot gun" is defined as a machine gun which has fired at least 200 rounds within a 2-minute period.)

a. Do not open cover.
b. Attempt to charge and fire the machine gun.
c. If firing cannot be resumed, retract the breech bolt to the seared position if possible, position the safety in safe position, and wait 30 minutes before attempting to clear machine gun and continuing firing.

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4.6.8.7 No parts shall be altered and only parts broken or worn to the extent that they are unserviceable shall be replaced.

4.6.8.8 Complete accurate records shall be kept for each endurance test, showing each malfunction and part replacement including the number of the round at which each occurred.

4.6.8.9 After completion of the endurance test, disposition of endurance tested machine guns shall be as specified in the contract (see 6.1).

4.6.9 Interchange of parts.

4.6.9.1 In plant.

4.6.9.1.1 Machine guns. Machine guns shall be tested for interchange of parts (see 3.4) by disassembling and then reassembling parts using the parts and prearranged system specified below. Interchange of parts shall be accomplished by dividing the parts of each machine gun into 10 groups of non-mating parts as shown below and distributing the groups into 10 different trays until each tray contains parts for a complete machine gun. Groups of parts from the first machine gun shall be taken in order and placed in trays 1 through 10, groups of parts from the second machine gun shall be taken in order and placed in trays 2 through 10 to 1; groups of parts from the third machine gun shall be taken in order and placed in trays 3 through 10 to 2; etc. Commercial parts such as screws, spring pins, etc., shall be placed in the same tray as their mating or associate part. Any commercial part rendered unserviceable by disassembly shall be replaced without penalty to the interchangeability test. The machine guns shall be reassembled using only those parts which are in the same tray.

4.6.9.1.2 Concurrent repair parts. Concurrent repair parts shall be tested for interchangeability requirement (see 3.4) by disassembling two machine guns, previously tested in 4.5.3.9.1.1, as necessary and then reassembling them using the concurrent repair parts. No hand refinement of parts will be allowed, and the machine guns shall operate and function properly. This test may be performed independently of the machine gun interchangeability test specified in 4.5.3.9.1.1 and at more frequent intervals using accepted machine guns taken from current production.

4.6.9.2 Interplant. Machine guns to be subjected to the interplant interchangeability test shall be given preliminary hand functioning to assure proper operation before parts are disassembled. In addition, the machine guns shall be inspected for solenoid gap adjustment, firing pin protrusion, sear release, breech bolt gap, headspace, accuracy and functioning firing before and after interchange of parts using the inspection methods specified in 4.5.2.2.16 d, 4.6.1, 4.6.2, 4.6.3, 4.6.4, 4.6.6 and 4.6.7, respectively. Readjustment of the solenoid gap shall be allowed after interchanging of parts. Machine guns shall be interchanged in a manner similar to the detailed plan specified in 4.6.9.1.1 except that parts shall be divided into six groups and when disassembling, every other machine gun used shall be one produced by a different manufacturer. Before machine guns are returned to the contractors, the original parts shall be reassembled to their respective machine guns, and the machine guns shall be hand functioned to assure proper operation.

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Groups of non-mating parts

Group I

Gun, latch (7793083)
Extension, subassembly barrel (7791371) (with Interlock, barrel extension-7791278, and Spring, helical compression-7791265 removed)
Guide assembly, link (11010134)
Latch, back plate (8448210)
Pin, firing (7791243)
Pin, quick release (7793046)
Pin, straight, headless (7790894)
Plunger, accelerator (7790981)
Sleeve, buffer (7791185)
Spring, helical, compression (7791411)
Spring, helical, compression (7793203) (with Insert, spring-7791109-2)
Spring, helical, torsion (7793114)
Tray, feed (7793049)

Group II

Barrel with flash suppressor (8448439)
Feed assembly, actuator (7793098)
Lever assembly, latch (7793211)
Pawl, belt retaining (8448226)
Pin, guide spring (7793239)
Pin, quick release (7793077)
Pin, straight, headless (7791183)
Slide, breechblock (7791383)
Spring, helical, compression (7793126)
Spring, helical, compression (7793215)
Spring, helical, compression (7793224)
Striker assembly (7791361)
Trigger assembly (11010179)

Group III

Cover subassembly (7793150)
Extractor, small arms cartridge (7793193)
Housing, return, feed (7793225)
Lock assembly, barrel latch (7791078)
Pin, quick release (7790905)
Pin, stripper, retaining (7793236)
Retainer, cartridge case (7791378)
Rod, return, barrel (7791370)
Solenoid (7792921)
Spring, helical, compression (7793106)
Spring, helical, compression (7793250)
Spring, helical, torsion (8228 228)
Tube assembly, charger (7793025)

Group IV

Detent assembly, charger (7793178)
Disk, rubber (7793172)
Pawl, cartridge guide (7793244)
Pin, extractor (7791239)
Pin, straight, headless (7791077)
Receiver assembly (7793062)
Sear, bolt (7793249)
Selector, rate (7793074)
Slide, feed (8228227)
Spring, helical, compression (7791351)
Spring, helical, compression (7791265)
Spring, helical, compression (7793205) (with Insert, springs-7791109-1)
Washer, thrust (7793216)

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Group V

Accelerator (7790977)
Disk, center (7793147)
Latch assembly, barrel (7790984)
Lever, belt feed (8228225)
Pin, belt retaining pawl (7793102)
Pin, lock (7790883)
Pin, straight, headed (7793243)
Pin, straight, headless (7793112)
Retainer, rate selector (7793078
Spacer, sleeve (7793075)
Spring, helical, compression (7790739)
Spring, helical, compression (7793235)
Spring, helical, torsion (7793251)
Support, feed and ejector (7793109)

Group VI

Breechblock (7791382)
Disk, rear (7793188)
Knob, detent (7793180)
Latch, cover (7793230)
Pawl, cartridge stop (7793222)
Pin, quick release (7793044)
Plate, back (7791381 or 7791385)
Rod, guide (7791467)
Spring, helical, compression (7793047)
Spring, helical, compression (7793183)
Spring, helical, torsion (7793252)
Stripper, cartridge link (7793234)

Group VII

Cylinder, time delay (7793076)
Disk, front (7793163)
Guide, feed slide (7793233)
Lever, bolt sear (7793245)
Pin, accelerator (7790979)
Pin, straight, headed (7793196)
Plunger, retainer (7793104)
Ramp, guide, cartridge (7793232)
Spring, helical, compression (7793092)
Spring, helical, compression (7793137)
Spring, helical, compression (7793206) (with insert, spring-7791109-3)
Spring, helical, compression (7793238)

Group VIII

Block, sear (7793194)
Extension assembly, trigger (11686373)
Interlock, barrel extension (7791278)
Lock, breechblock, LH (7791181)
Pawl, belt feed (8228235)
Pin, quick release (7793045)
Pin, retaining (7791057)
Retainer, pin (7793031)
Spring, helical, compression (7790982)
Spring, helical, compression (7790983)
Spring, helical, compression (7793228)
Spring, helical, compression (7793242)

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Group IX

Detent, cover (7793132)
Disconnector assembly (7793088)
Handle assembly, charger (11010488)
Housing, buffer (7793248)
Lever assembly, feed actrator (7791450)
Pin, bolt retracting (7793185)
Pin, quick release (7790849)
Plunger, cartridge stop pawl (7793223)
Rammer, cartridge (7791374)
Safety assembly (7793258)
Spring, helical, compression (7791254)
Spring, helical, compression (7793108)
Spring, torsion (7793058)
Detent, sear block (7793195)
Ejector, small arms cartridge (7791375)
Housing, accelerator (7790978)
Lock, breechblock, R H (7791182)
Lock, latch (8448225)
Pin, straight, headed (7791350)
Pin, straight, headless (7793226)
Pin, straight, headless (7793246)
Retainer, feed assembly (7793099)
Shaft, cylinder (7793079)
Spring, helical, compression (7793105)
Spring, helical, compression (7793184)

5. PREPARATION FOR DELIVERY

* 5.1 Pilot pack. The pilot pack shall consist of a complete machine gun packaged in accordance with Packaging Data Sheet P12002938. Pilot packs shall be packaged to the level specified in the contract (see 6.1), packed level B and forwarded in accordance with 3.1.

* 5.2 Level A and B. Machine guns shall be preserved, packaged, packed, and marked in accordance with Packaging Data Sheet P12002938 for the applicable level of protection specified (see 6.1).

* 5.3 Commercial. Preservation, packaging, packing and marking shall be in accordance with Fed. Std. No. 356.

* 5.4 Repair parts. Repair parts shall be prepared for delivery in accordance with the applicable packaging data sheets or other requirements for the level of protection specified in the contract (see 6.1).

6. NOTES.

* 6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Lists of drawings and specifications pertinent to the machine gun, showing applicable revision dates.
c. Shipping instructions for first article and pilot pack (see 3.1, 4.4 and 5.1).

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d. Block of serial numbers (see 3.5).
e. Inspection criteria for components (see 4.5.2.1 and 4.5.3.11).
f. QE-G-DL Data Lists pertinent to the machine gun, showing applicable revision dates (4.5.5.1).
g. List of acceptance inspection equipment to be furnished the contractor (see 4.5.5.1) and responsibilities for other Government property to be furnished the contractor.
h. Shipping instructions for machine guns and parts when an interplant interchangeability test is required (see 4.5.3.9.2).
i. Disposition of endurance tested guns (see 4.6.8.9).
j. Selection of applicable levels of preservation, packaging, packing and marking (see 5.1, 5.2 and 5.3).
k. Packaging instructions for repair parts (see 5.4).

6.2 Supersession data. The basic edition of this specification included the requirements of Springfield Armory Purchase Description SAPD-186A, dated 1 April 1966.

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions), from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notation and relationship to the last previous issue.

Custodian:
Army - WC
Navy - OS

Review activity:
Navy - OS

User activity:
Navy - MC

Preparing activity:
Army - WC

Project number:
1005-0537

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POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Rock Island Arsenal
ATTN: SARRI-LE-S
Rock Island, Illinois 61201

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-M 45590A; Machine Gun, Caliber .50: Fixed, M85

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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